

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER– VI (NEW) EXAMINATION – WINTER 2021****Subject Code:3161003****Date:24/11/2021****Subject Name:Antennas and Propagation****Time:10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- Q.1** (a) Define and discuss the Directivity of antenna under test with help of necessary formula. **03**
- (b) Define the following terms: **04**
- i) Radiation efficiency
 - ii) HPBW
 - iii) Beam area
 - iv) Radiation pattern
- (c) Explain different types of antenna with necessary figures. **07**
- Q.2** (a) "Antenna is a transitional structure". Justify **03**
- (b) Discuss the different feeding methods with necessary figures. **04**
- (c) Discuss the different types of antenna aperture with necessary figures and equations. **07**
- OR**
- (c) Derive the expression for the radiation resistance of half-wave dipole. **07**
- Q.3** (a) Explain half power beam width with respect to antenna field and power pattern. **03**
- (b) Sketch the helical geometry with its associated dimensions showing relationship between circumference, spacing, turn length and pitch angle of helix. **04**
- (c) Explain Pattern Multiplication and show that it can be used to find the resultant pattern of a linear array. Specify the limitations of this method. **07**
- OR**
- Q.3** (a) Define axial ratio. What is the ideal value of it for circular polarization? **03**
- (b) Discuss the antenna field zone with necessary figure. **04**
- (c) Write the equation of Scalar potential and vector potential for short dipole and derive the equation of electric field components and magnetic field components from that. **07**
- Q.4** (a) The radiation resistance of antenna is 70Ω and loss resistance is 10Ω . Calculate antenna efficiency. **03**
- (b) Explain about ultra-wideband antenna. **04**
- (c) Draw and explain structure of Log periodic antenna. Derive its mathematical equations for designing purpose. **07**

OR

- Q.4** (a) Calculate the gain for parabolic antenna with circular aperture of diameter 6 meter at a frequency of 100 MHz. **03**
(b) List out the uses of loop antenna. **04**
(c) Explain (i) Binomial arrays. **07**
(ii) Schelkunoff theorems for linear arrays.

- Q.5** (a) Explain the features of Yagi Uda antenna. **03**
(b) Explain the different types of lens antenna? **04**
(c) Explain the operation of Microstrip patch antenna with neat diagram. **07**

OR

- Q.5** (a) Discuss significance of complementary antenna with the help of babinet's principal. **03**
(b) Define the following terms.(draw necessary figures) **04**
(i) Duct propagation (ii) Skip zone
(c) Enlist and explain the Different modes of Radio wave propagation. **07**
